

Hadiri — System Design Document

Versi: 1.0

Stack: React PWA · Node.js/Express · PostgreSQL

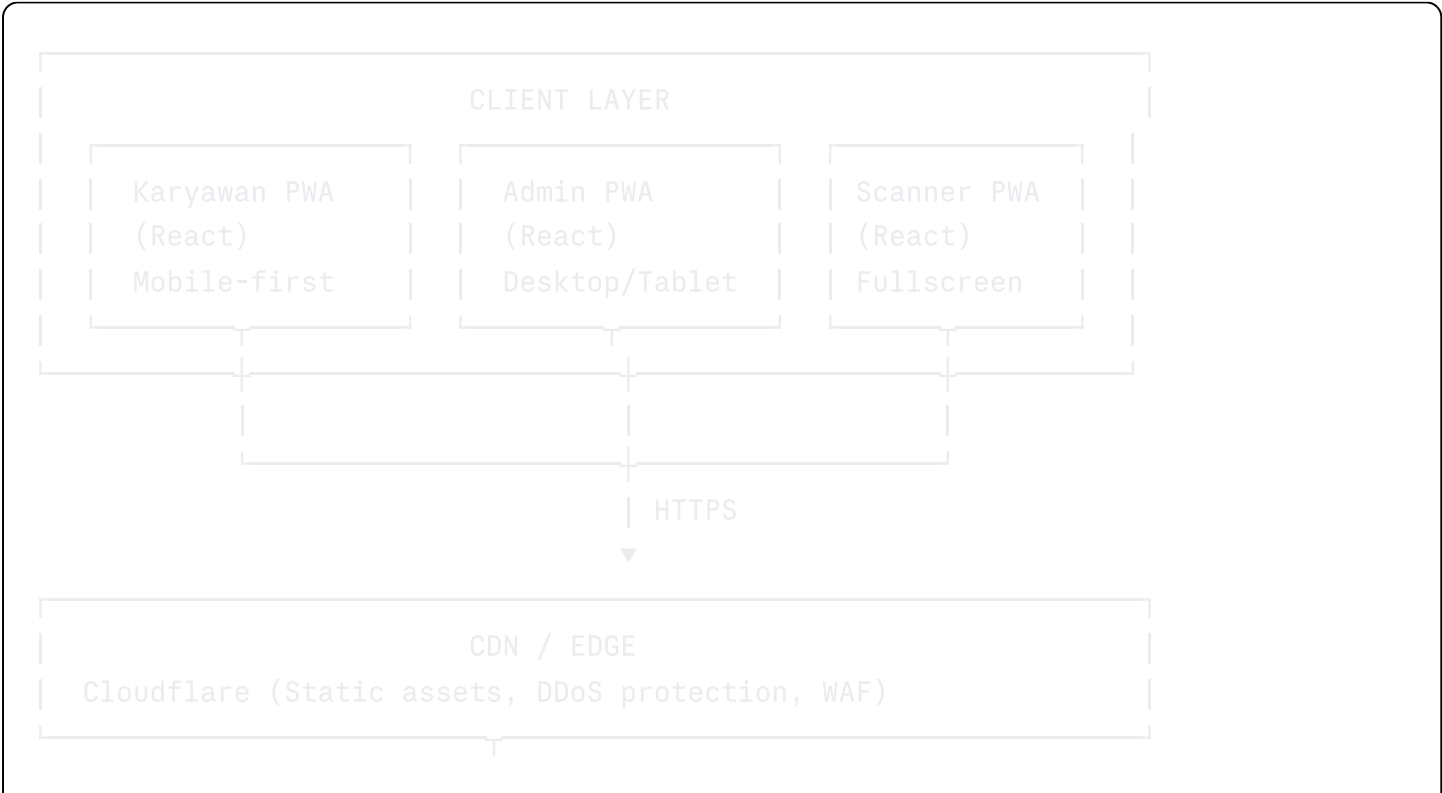
Update: April 2026

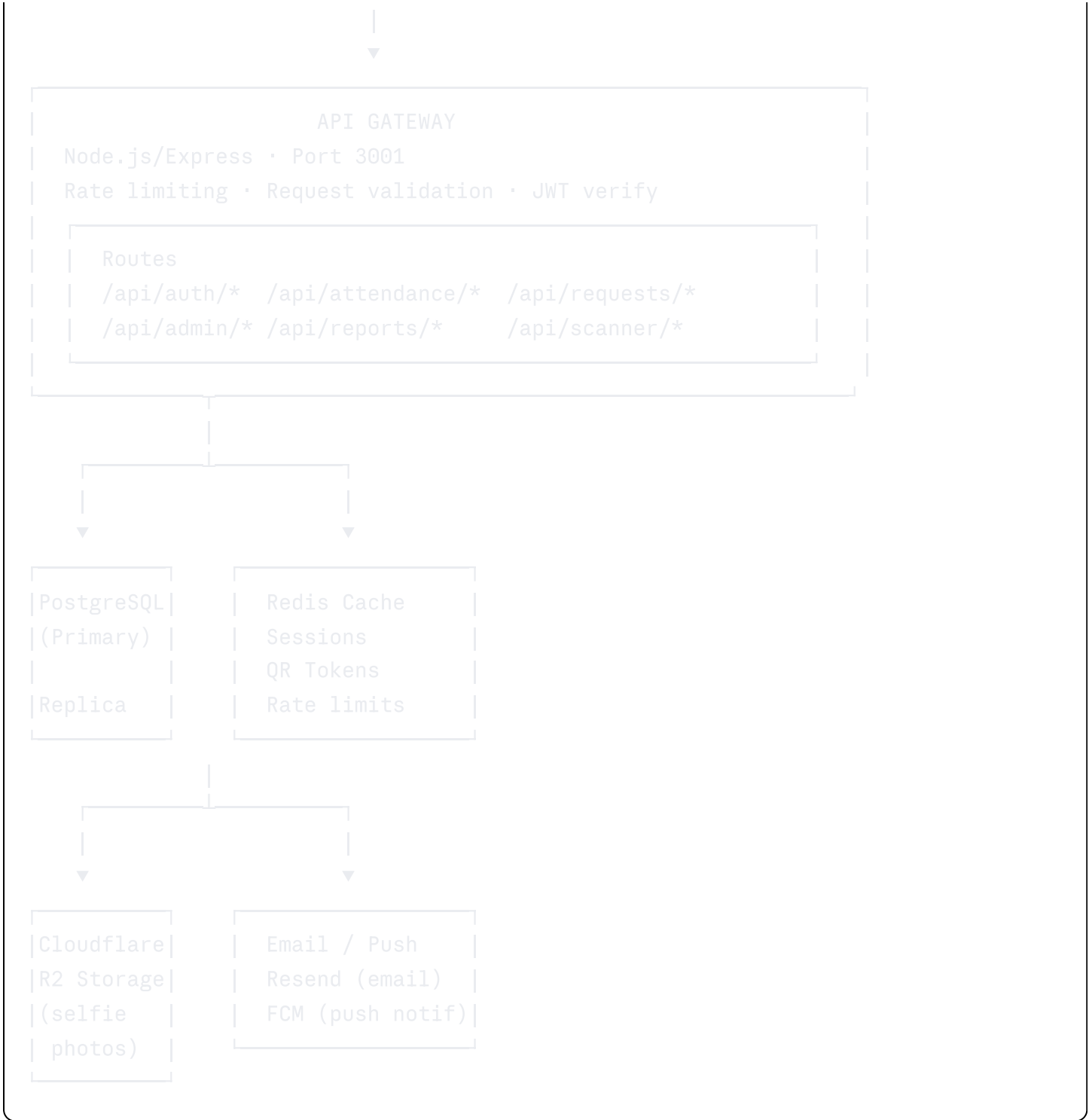
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1. Arsitektur Sistem Overview

High-Level Architecture





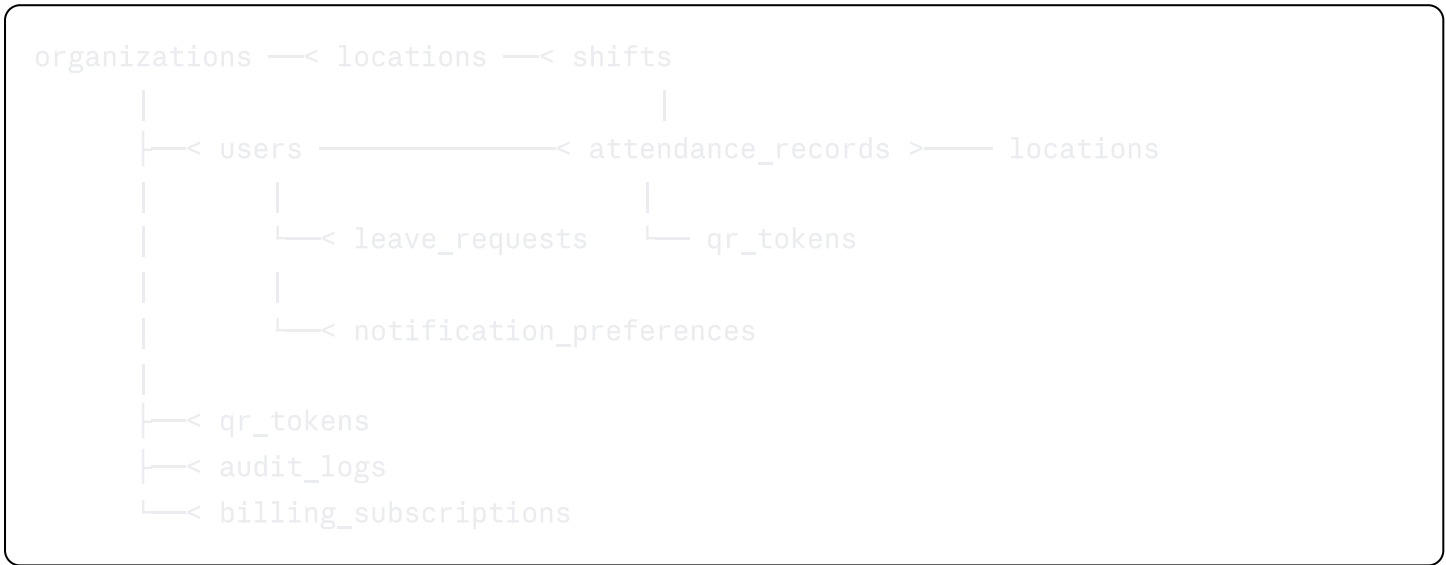
Komponen Utama

Komponen	Teknologi	Fungsi
Frontend PWA	React 18, Vite, TailwindCSS	UI semua role
Backend API	Node.js 20, Express 5	Business logic, API
Database	PostgreSQL 16	Data utama
Cache	Redis 7	Session, QR token, rate limit
File Storage	Cloudflare R2	Foto selfie, lampiran

Komponen	Teknologi	Fungsi
Email	Resend	OTP, notifikasi, invite
Push Notif	Firebase FCM	Mobile push notification
CDN/Proxy	Cloudflare	Static assets, protection
Monitoring	Grafana + Prometheus	Metrics, alerting

2. Database Design (ERD)

Entity Relationship Diagram (Teks)



Tabel: organizations

Kolom	Type	Constraint	Keterangan
id	UUID	PK, DEFAULT gen_random_uuid()	
name	VARCHAR(255)	NOT NULL	Nama organisasi
type	ENUM	NOT NULL	'corporate','school','ukm','startup'
slug	VARCHAR(100)	UNIQUE, NOT NULL	URL-friendly, auto-generated
logo_url	TEXT	NULL	URL di Cloudflare R2
timezone	VARCHAR(50)	DEFAULT 'Asia/Jakarta'	

Kolom	Type	Constraint	Keterangan
default_work_start	TIME	DEFAULT '08:00'	
default_work_end	TIME	DEFAULT '17:00'	
late_tolerance_minutes	INTEGER	DEFAULT 15	
plan	ENUM	DEFAULT 'free'	'free','starter','pro','enterprise'
max_employees	INTEGER	DEFAULT 10	Berdasarkan plan
is_active	BOOLEAN	DEFAULT true	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: slug, plan

Tabel: users

Kolom	Type	Constraint	Keterangan
id	UUID	PK	
organization_id	UUID	FK organizations, NOT NULL	
email	VARCHAR(255)	UNIQUE, NOT NULL	
password_hash	VARCHAR(255)	NOT NULL	bcrypt, cost 12
full_name	VARCHAR(255)	NOT NULL	
phone	VARCHAR(20)	NULL	
role	ENUM	NOT NULL	'superadmin','admin','employee'
department	VARCHAR(100)	NULL	
employee_id	VARCHAR(50)	NULL	ID karyawan eksternal
avatar_url	TEXT	NULL	
shift_id	UUID	FK shifts, NULL	Shift default
annual_leave_quota	INTEGER	DEFAULT 12	

Kolom	Type	Constraint	Keterangan
annual_leave_used	INTEGER	DEFAULT 0	FCM token
push_token	TEXT	NULL	
email_notif	BOOLEAN	DEFAULT true	
push_notif	BOOLEAN	DEFAULT true	
is_active	BOOLEAN	DEFAULT true	
last_login_at	TIMESTAMPTZ	NULL	
invited_by	UUID	FK users, NULL	
invite_accepted_at	TIMESTAMPTZ	NULL	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: organization_id, email, role, shift_id

Tabel: locations

Kolom	Type	Constraint	Keterangan
id	UUID	PK	"Kantor Pusat"
organization_id	UUID	FK organizations, NOT NULL	
name	VARCHAR(255)	NOT NULL	
address	TEXT	NULL	
latitude	DECIMAL(10,8)	NOT NULL	
longitude	DECIMAL(11,8)	NOT NULL	
radius_meters	INTEGER	DEFAULT 200	
is_active	BOOLEAN	DEFAULT true	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: organization_id

Tabel: shifts

Kolom	Type	Constraint	Keterangan
id	UUID	PK	
organization_id	UUID	FK organizations, NOT NULL	
name	VARCHAR(100)	NOT NULL	"Shift Pagi"
work_days	INTEGER[]	NOT NULL	[1,2,3,4,5] = Senin-Jumat
start_time	TIME	NOT NULL	
end_time	TIME	NOT NULL	
late_tolerance_minutes	INTEGER	NULL	Override dari org default
location_id	UUID	FK locations, NULL	Lokasi default shift ini
is_active	BOOLEAN	DEFAULT true	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Tabel: attendance_records

Kolom	Type	Constraint	Keterangan
id	UUID	PK	
user_id	UUID	FK users, NOT NULL	
organization_id	UUID	FK organizations, NOT NULL	Denormalized untuk query cepat
date	DATE	NOT NULL	Tanggal absensi
check_in_at	TIMESTAMPTZ	NULL	
check_out_at	TIMESTAMPTZ	NULL	
check_in_method	ENUM	NULL	'qr','selfie','gps','manual'

Kolom	Type	Constraint	Keterangan
check_out_method	ENUM	NULL	'qr','selfie','gps','manual'
check_in_location_id	UUID	FK locations, NULL	
check_out_location_id	UUID	FK locations, NULL	
check_in_lat	DECIMAL(10,8)	NULL	
check_in_lng	DECIMAL(11,8)	NULL	
check_out_lat	DECIMAL(10,8)	NULL	
check_out_lng	DECIMAL(11,8)	NULL	
selfie_url	TEXT	NULL	URL foto di R2
status	ENUM	NOT NULL	'present','late','absent','izin','cuti'
is_late	BOOLEAN	DEFAULT false	
late_minutes	INTEGER	NULL	
shift_id	UUID	FK shifts, NULL	Snapshot shift saat itu
work_duration_minutes	INTEGER	NULL	check_out - check_in
is_synced	BOOLEAN	DEFAULT true	false jika dari offline queue
offline_submitted_at	TIMESTAMP TZ	NULL	
notes	TEXT	NULL	Catatan admin/manual
created_at	TIMESTAMP TZ	DEFAULT NOW()	
updated_at	TIMESTAMP TZ	DEFAULT NOW()	

Indexes: ((user_id, date) UNIQUE, ((organization_id, date), (status), (user_id, created_at DESC))

Constraint: UNIQUE (user_id, date) — satu record per orang per hari

Tabel: `qr_tokens`

Kolom	Type	Constraint	Keterangan
<code>id</code>	UUID	PK	
<code>organization_id</code>	UUID	FK organizations, NOT NULL	
<code>location_id</code>	UUID	FK locations, NOT NULL	
<code>token</code>	VARCHAR(255)	NOT NULL	HMAC-SHA256 hash
<code>payload</code>	JSONB	NOT NULL	{orgId, locationId, timestamp, nonce}
<code>expires_at</code>	TIMESTAMPTZ	NOT NULL	NOW() + 30 seconds
<code>is_used</code>	BOOLEAN	DEFAULT false	
<code>used_by</code>	UUID	FK users, NULL	
<code>used_at</code>	TIMESTAMPTZ	NULL	
<code>created_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

Note: Token juga di-cache di Redis dengan TTL 35 detik

Indexes: `(token)`, `((organization_id, expires_at))`, `((location_id, expires_at))`

Cleanup: Cron job setiap jam hapus expired tokens

Tabel: `leave_requests`

Kolom	Type	Constraint	Keterangan
<code>id</code>	UUID	PK	
<code>user_id</code>	UUID	FK users, NOT NULL	
<code>organization_id</code>	UUID	FK organizations, NOT NULL	
<code>type</code>	ENUM	NOT NULL	'izin','cuti','koreksi'
<code>start_date</code>	DATE	NOT NULL	
<code>end_date</code>	DATE	NOT NULL	

Kolom	Type	Constraint	Keterangan
duration_days	INTEGER	NOT NULL	Hari kerja efektif
category	VARCHAR(100)	NULL	'sakit','pribadi','dinas_luar', dll
reason	TEXT	NOT NULL	
attachment_url	TEXT	NULL	
status	ENUM	DEFAULT 'pending'	'pending','approved','rejected'
reviewed_by	UUID	FK users, NULL	Admin yang review
reviewed_at	TIMESTAMPTZ	NULL	
review_notes	TEXT	NULL	Catatan approval/rejection
correction_date	DATE	NULL	Khusus type='koreksi'
correction_checkin	TIME	NULL	
correction_checkout	TIME	NULL	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: (user_id, status), (organization_id, status), (start_date, end_date)

Tabel: invite_tokens

Kolom	Type	Constraint	Keterangan
id	UUID	PK	
organization_id	UUID	FK organizations, NOT NULL	
email	VARCHAR(255)	NOT NULL	
role	ENUM	NOT NULL	'admin','employee'
shift_id	UUID	FK shifts, NULL	
token	VARCHAR(255)	UNIQUE, NOT NULL	Random secure token
invited_by	UUID	FK users, NOT NULL	
expires_at	TIMESTAMPTZ	NOT NULL	+7 hari

Kolom	Type	Constraint	Keterangan
accepted_at	TIMESTAMPTZ	NULL	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Tabel: audit_logs

Kolom	Type	Constraint	Keterangan
id	BIGSERIAL	PK	
organization_id	UUID	FK organizations, NOT NULL	
actor_id	UUID	FK users, NULL	NULL = sistem
action	VARCHAR(100)	NOT NULL	'checkin','approve_request', dll
entity_type	VARCHAR(50)	NOT NULL	'attendance','leave_request', dll
entity_id	UUID	NOT NULL	
old_values	JSONB	NULL	Snapshot sebelum
new_values	JSONB	NULL	Snapshot sesudah
ip_address	INET	NULL	
user_agent	TEXT	NULL	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Retention: 2 tahun, lalu archive ke cold storage

Indexes: (organization_id, created_at), (entity_type, entity_id), actor_id

Tabel: billing_subscriptions

Kolom	Type	Constraint	Keterangan
id	UUID	PK	

Kolom	Type	Constraint	Keterangan
<code>organization_id</code>	UUID	FK organizations, UNIQUE	
<code>plan</code>	ENUM	NOT NULL	'free','starter','pro','enterprise'
<code>price_idr</code>	INTEGER	NOT NULL	Dalam Rupiah
<code>billing_cycle</code>	ENUM	DEFAULT 'monthly'	'monthly','yearly'
<code>current_period_start</code>	DATE	NOT NULL	
<code>current_period_end</code>	DATE	NOT NULL	
<code>status</code>	ENUM	NOT NULL	'active','past_due','cancelled'
<code>payment_method</code>	VARCHAR(50)	NULL	
<code>external_subscription_id</code>	VARCHAR(255)	NULL	ID dari payment gateway
<code>cancelled_at</code>	TIMESTAMPTZ	NULL	
<code>created_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

3. API Design

Base URL

```
Production:  https://api.hadiri.app/api/v1
Staging:    https://api-staging.hadiri.app/api/v1
```

Authentication

Semua endpoint (kecuali `/auth/*`) memerlukan:

```
Authorization: Bearer {jwt_access_token}
```

Conventions

- Response format: JSON
- Success: `{ data: {...}, meta: {...} }`
- Error: `{ error: { code, message, details } }`

- Timestamp: ISO 8601 UTC
 - Pagination: `?page=1&limit=25`
-

Auth Endpoints

POST `/auth/register`

json

Request:

```
{
  "organizationName": "PT. Maju Bersama",
  "email": "admin@perusahaan.com",
  "password": "Password123!",
  "phone": "08123456789",
  "organizationType": "corporate"
}
```

Response 201:

```
{
  "data": {
    "userId": "uuid",
    "organizationId": "uuid",
    "message": "OTP sent to email"
  }
}
```

POST `/auth/verify-otp`

json

Request:

```
{ "email": "admin@perusahaan.com", "otp": "123456" }
```

Response 200:

```
{ "data": { "accessToken": "...", "refreshToken": "..." } }
```

POST `/auth/login`

json

Request:

```
{ "email": "...", "password": "..." }
```

Response 200:

```
{ "data": { "accessToken": "...", "refreshToken": "...", "user": {...}, "organization": {...} } }
```

POST `/auth/refresh`

json

```
Request: { "refreshToken": "..." }
```

```
Response 200: { "data": { "accessToken": "..." } }
```

POST `/auth/logout` — Invalidate refresh token

POST `/auth/forgot-password` — Kirim email reset

POST `/auth/reset-password` — Token dari email + password baru

Attendance Endpoints

POST `/attendance/checkin`

json

Request (QR):

```
{
  "method": "qr",
  "qrToken": "hmac-token-string",
  "latitude": -7.0051,
  "longitude": 110.4381
}
```

Request (Selfie):

```
{
  "method": "selfie",
  "photo": "base64...",
  "latitude": -7.0051,
  "longitude": 110.4381
}
```

Request (GPS):

```
{
  "method": "gps",
  "latitude": -7.0051,
  "longitude": 110.4381,
  "accuracy": 8.5
}
```

Response 201:

```
{
  "data": {
    "attendanceId": "uuid",
    "checkInAt": "2026-04-28T01:02:00Z",
    "status": "present",
    "isLate": false,
    "location": { "id": "...", "name": "Kantor Pusat" },
    "selfieUrl": null
  }
}
```

POST `/attendance/checkout`

json

```
Request: { "latitude": -7.0051, "longitude": 110.4381 }
Response 200: { "data": { "checkOutAt": "...", "workDurationMinutes": 542 } }
```

GET `/attendance/today` — Status kehadiran hari ini user yang login

GET `/attendance/history`

```
Query: ?month=2026-04&page=1&limit=20
```

```
Response: { data: [...records], meta: { total, page, limit, summary: {present, late, absent} } }
```

GET `/attendance/:id` — Detail satu record

Scanner Endpoints

POST `/scanner/auth` — Login scanner dengan PIN

json

```
Request: { "organizationSlug": "pt-maju-bersama", "locationId": "uuid", "pin": "1234"
```

```
Response: { "data": { "scannerToken": "...", "expiresAt": "..." } }
```

GET `/scanner/token` — Get QR token saat ini

json

```
Headers: Authorization: Bearer {scannerToken}
```

```
Response:
```

```
{  
  "data": {  
    "token": "hmac-sha256-string",  
    "expiresAt": "2026-04-28T01:41:30Z",  
    "location": { "id": "...", "name": "Kantor Pusat" }  
  }  
}
```

Leave Requests Endpoints

GET `/requests`

```
Query: ?type=izin|cuti|koreksi&status=pending|approved|rejected&page=1
```

POST `/requests`

json

Request:

```
{
  "type": "izin",
  "startDate": "2026-05-02",
  "endDate": "2026-05-02",
  "category": "sakit",
  "reason": "Demam tinggi",
  "attachmentUrl": "r2://..."
}
```

Response 201: { "data": { "requestId": "...", "status": "pending" } }

GET `/requests/:id` — Detail satu request

DELETE `/requests/:id` — Batalkan request (status pending only)

Admin Endpoints

GET `/admin/dashboard`

json

Response:

```
{
  "data": {
    "today": {
      "present": 41, "late": 4, "absent": 3,
      "total": 48, "notCheckedIn": [...]
    },
    "last30days": [
      { "date": "2026-04-01", "presentRate": 0.89 },
      ...
    ]
  }
}
```

GET `/admin/attendance` — Log kehadiran dengan filter + pagination

GET `/admin/requests` — List requests dengan filter pending/approved/rejected

PATCH `/admin/requests/:id`

json

Request: { "status": "approved" | "rejected", "reviewNotes": "..." }

GET `/admin/employees` — List karyawan dengan filter

POST `/admin/employees/invite`

json

Request: { "emails": ["..."], "role": "employee", "shiftId": "uuid" }

PATCH `/admin/employees/:id` — Update data karyawan

DELETE `/admin/employees/:id` — Nonaktifkan (soft delete)

GET `/admin/reports` — Laporan dengan filter

Query: ?startDate=2026-04-01&endDate=2026-04-30&employeeId=...&status=...

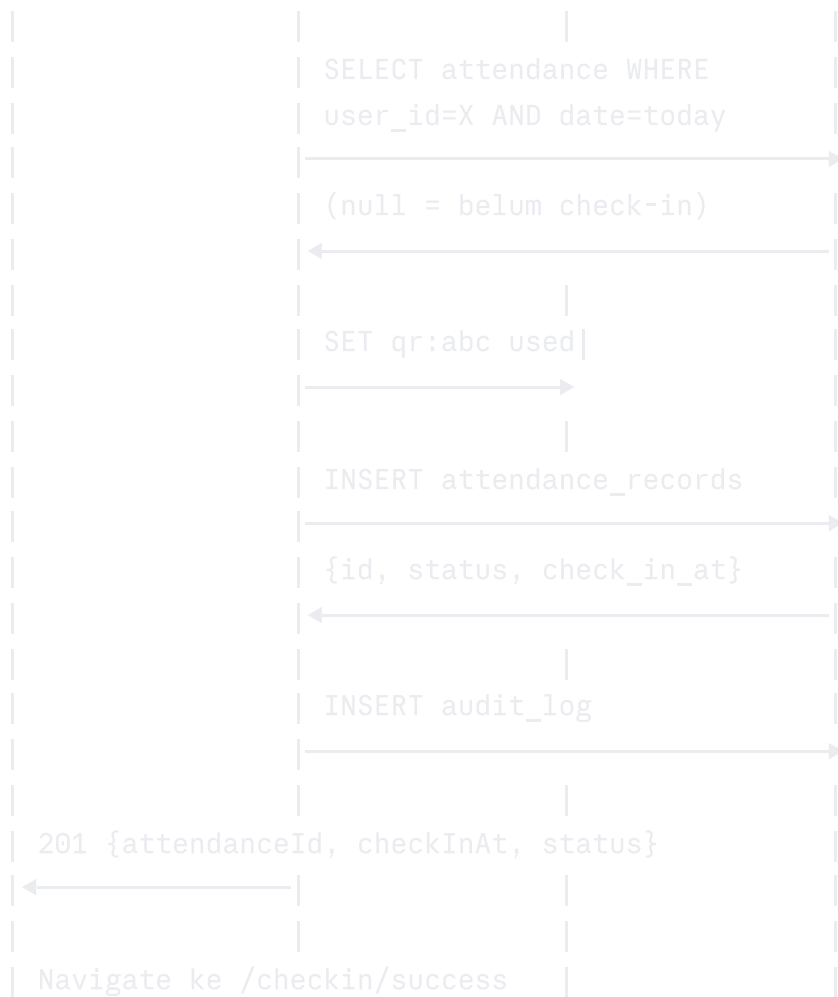
GET `/admin/reports/export`

Query: ?format=xlsx|pdf&...filters
Response: File download (Content-Disposition: attachment)

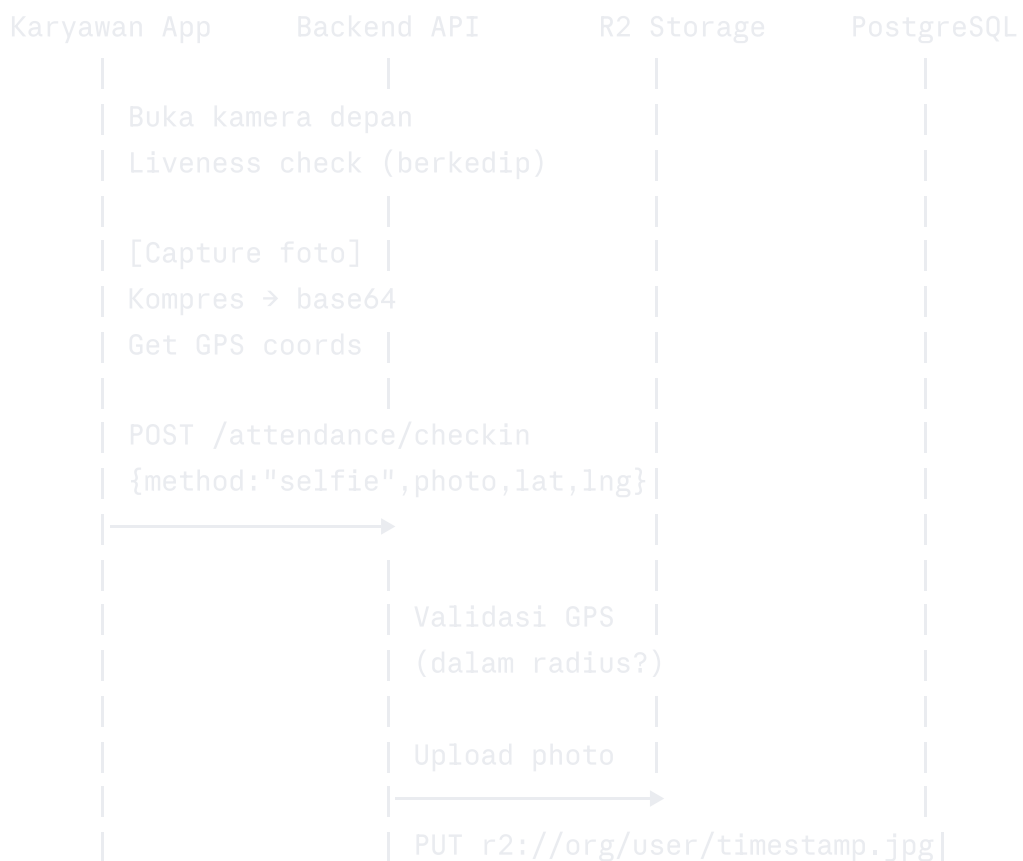
4. Sequence Diagrams

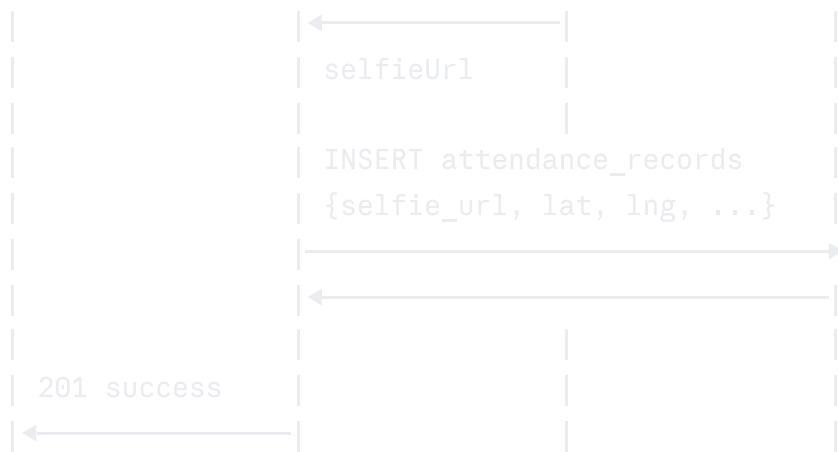
4.1 QR Check-in Flow



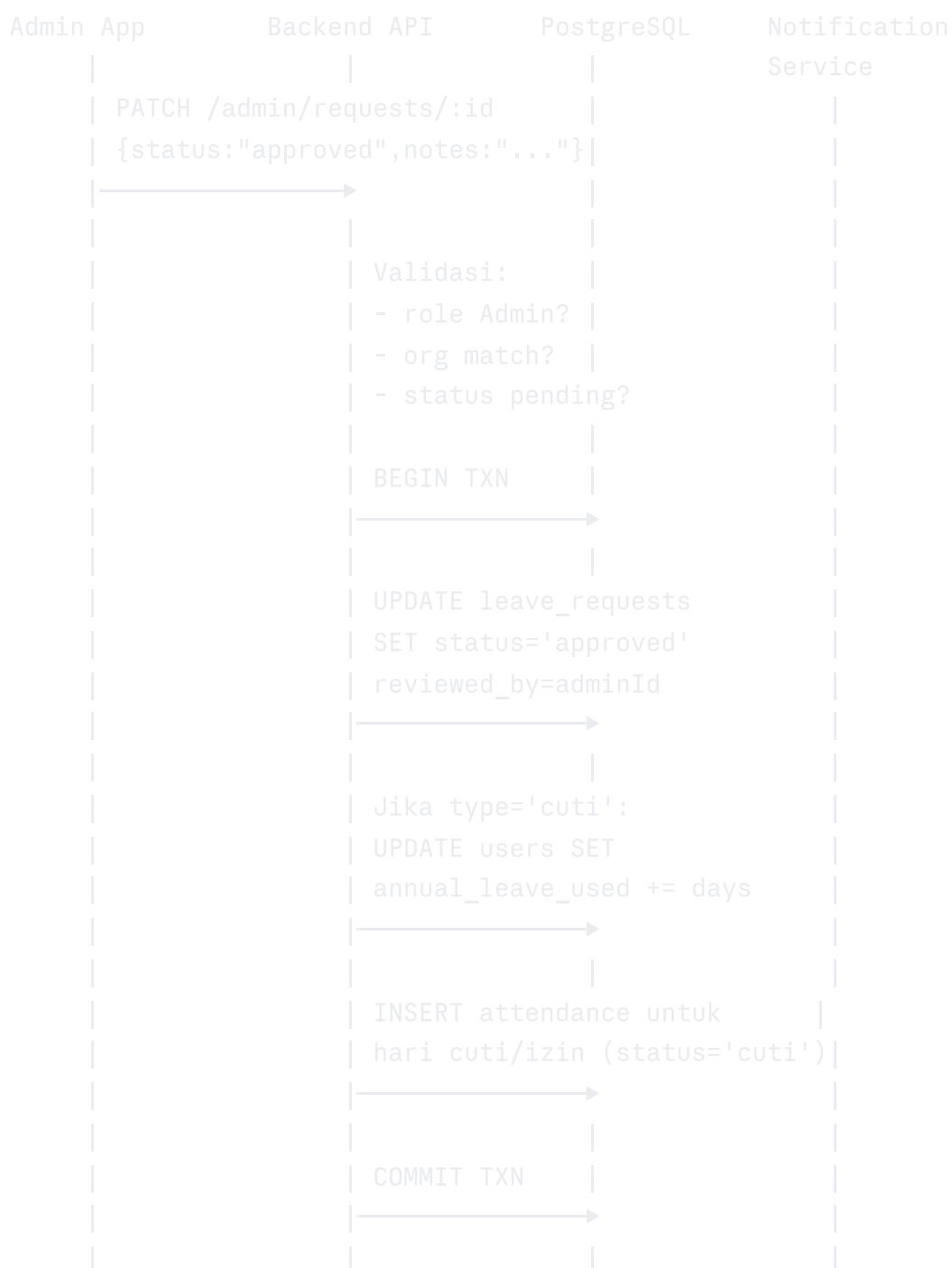


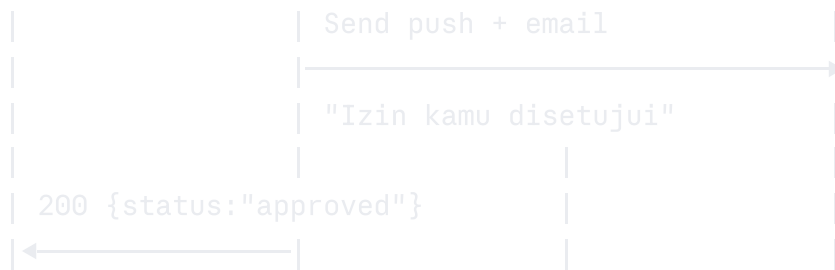
4.2 Selfie Check-in Flow



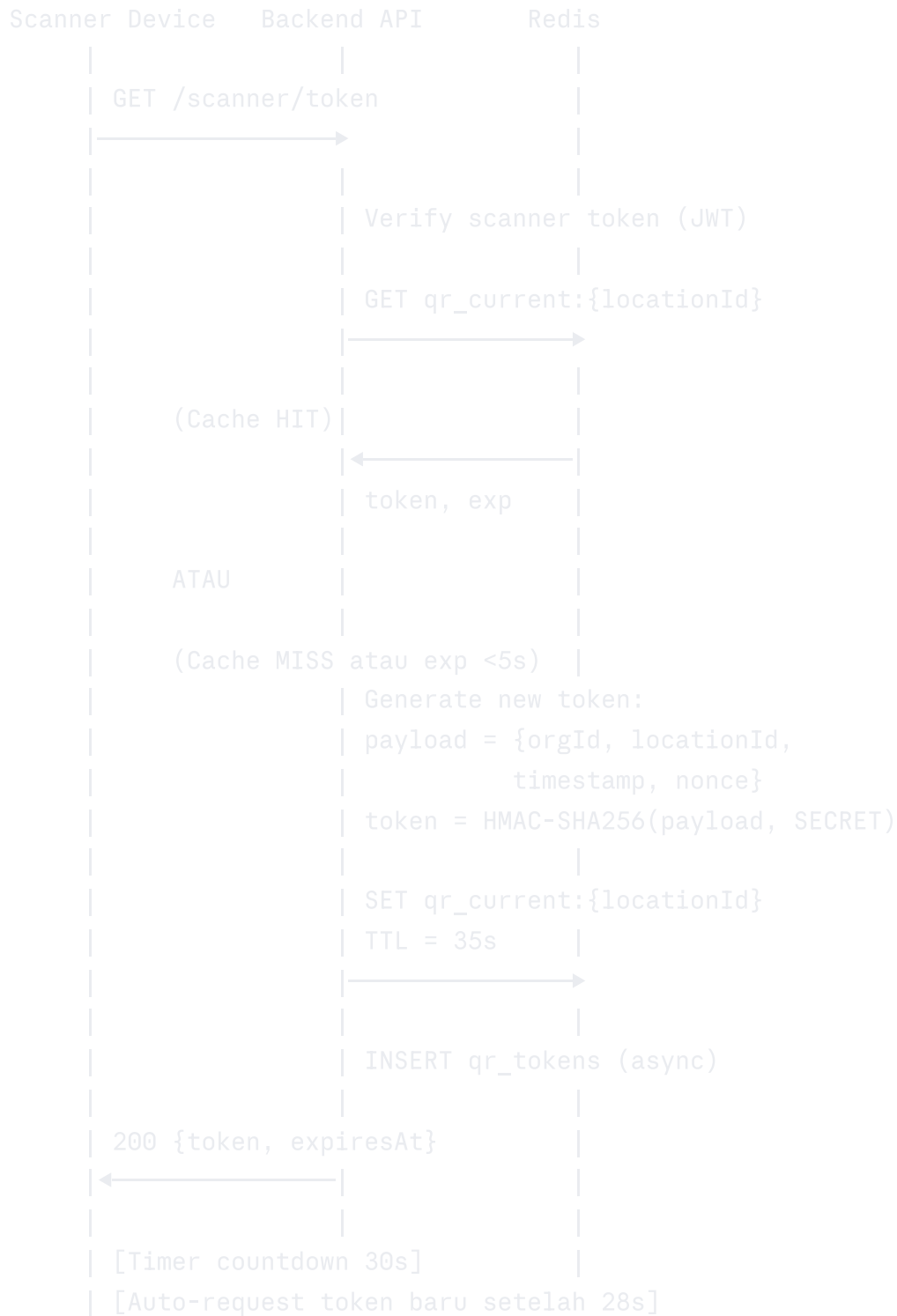


4.3 Admin Approve Request Flow





4.4 QR Token Generation (Scanner)



5. Infrastruktur & Deployment

Environment Structure



Environment Variables

Backend (.env):

```
env
```

```
# Server
NODE_ENV=production
PORT=3001
API_VERSION=v1

# Database
DATABASE_URL=postgresql://user:pass@host:5432/hadiri_prod
DATABASE_POOL_MIN=2
DATABASE_POOL_MAX=10

# Redis
REDIS_URL=redis://default:pass@host:6379

# JWT
JWT_ACCESS_SECRET=<256-bit random>
JWT_REFRESH_SECRET=<256-bit random>
JWT_ACCESS_EXPIRES=15m
JWT_REFRESH_EXPIRES=30d

# QR Token
QR_HMAC_SECRET=<256-bit random>
QR_TOKEN_TTL_SECONDS=30

# Storage
CLOUDFLARE_R2_ACCOUNT_ID=...
CLOUDFLARE_R2_ACCESS_KEY=...
CLOUDFLARE_R2_SECRET_KEY=...
CLOUDFLARE_R2_BUCKET=hadiri-prod
CLOUDFLARE_R2_PUBLIC_URL=https://files.hadiri.app

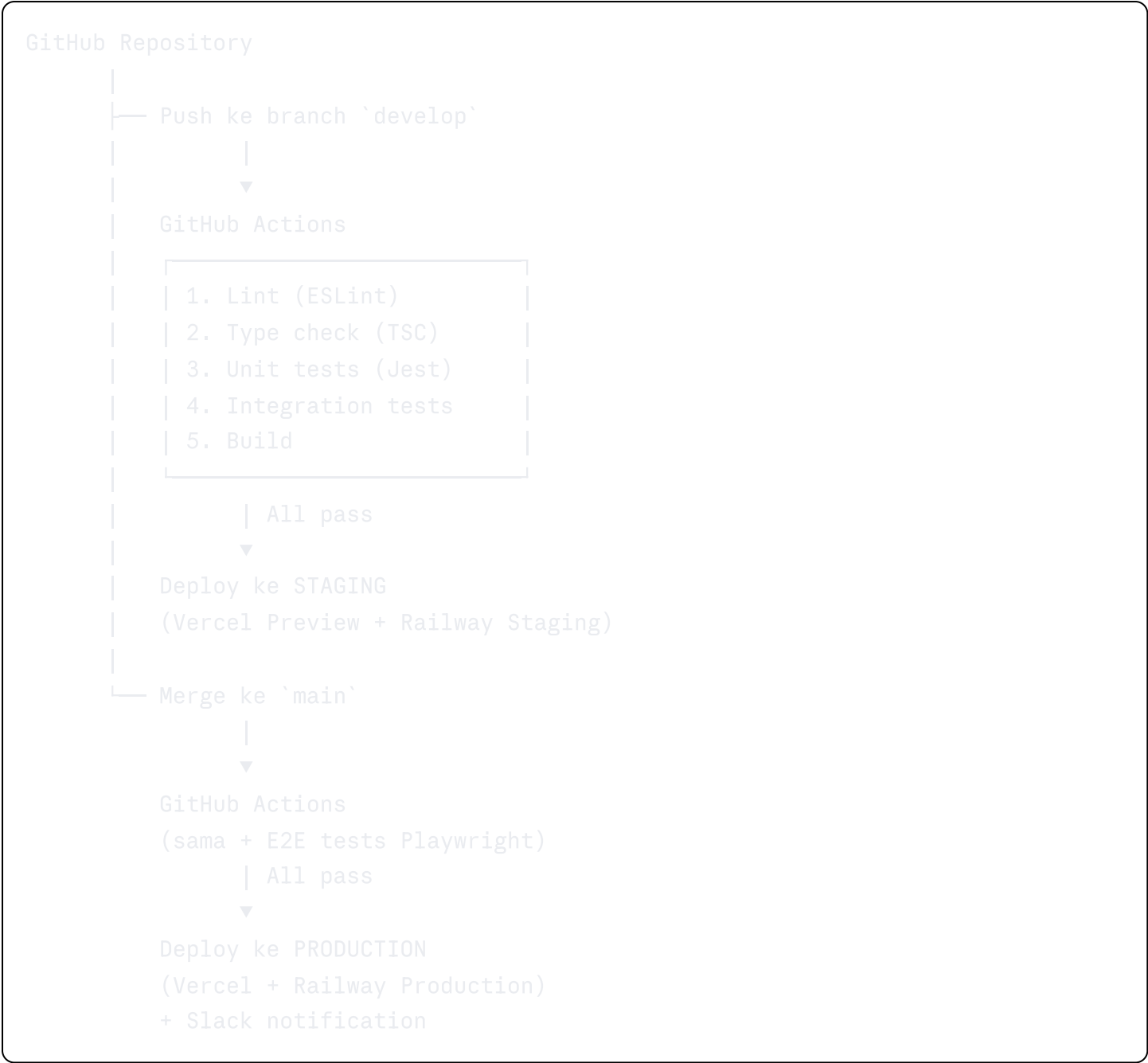
# Email
RESEND_API_KEY=re_...

# Push Notification
FCM_SERVICE_ACCOUNT_JSON=...

# Rate Limiting
RATE_LIMIT_WINDOW_MS=60000
RATE_LIMIT_MAX_REQUESTS=100
CHECKIN_RATE_LIMIT_MAX=3

# Security
CORS_ORIGINS=https://hadiri.app,https://admin.hadiri.app
BCRYPT_ROUNDS=12
```

Deployment Pipeline (CI/CD)



Scaling Strategy

Komponen	Phase 1 (0-1000 user)	Phase 2 (1000-10000)	Phase 3 (10000+)
Backend	Single instance	2-3 instances	Auto-scale
Database	Single Postgres	Read replica	Sharding per org
Cache	Upstash Redis	Redis cluster	Multi-region
Storage	R2 single bucket	R2 multi-region	CDN optimized
CDN	Cloudflare free	Cloudflare Pro	Enterprise

6. Security Design

Authentication & Authorization

JWT Flow:

Access Token: expire 15 menit → rotate agresif
Refresh Token: expire 30 hari → stored httpOnly cookie

Payload access token:

```
{
  "sub": "userId",
  "orgId": "organizationId",
  "role": "admin",
  "iat": timestamp,
  "exp": timestamp
}
```

Role-Based Access Control (RBAC):

Endpoint	Karyawan	Admin	Superadmin
/attendance/*	✓ (own)	✓ (all)	✓ (all)
/requests/*	✓ (own)	✓ (all)	✓ (all)
/admin/*	✗	✓	✓
/admin/settings	✗	Read	Write
/admin/employees/delete	✗	✗	✓
/admin/billing	✗	✗	✓

QR Token Security

Token generation:

- payload = JSON.stringify({
 orgId, locationId,
 timestamp: Date.now(),
 nonce: crypto.randomBytes(16).toString('hex')
})
- token = crypto.createHmac('sha256', QR_HMAC_SECRET)
 .update(payload)
 .digest('hex')
- Store di Redis: SET qr:{token} {payload} EX 35

Validation:

1. Decode token dari QR scan
2. GET dari Redis – kalau tidak ada: expired
3. Re-compute HMAC dan compare – kalau beda: invalid
4. Check expires_at – kalau lewat: expired
5. Check is_used – kalau true: already used
6. Mark as used: SET qr:{token}:used 1 EX 10

Anti-screenshot attack:

- Token expire 30 detik → screenshot tidak berguna dalam 30s
- Nonce unik per token → tidak bisa di-replay
- Mark used → tidak bisa dipakai dua kali

Rate Limiting

Endpoint	Limit	Window
POST /auth/login	5 req	5 menit per IP
POST /auth/register	3 req	1 jam per IP
POST /auth/verify-otp	3 req	10 menit per email
POST /attendance/checkin	3 req	1 menit per user
GET /scanner/token	10 req	1 menit per scanner
Global API	100 req	1 menit per user

Implementation: Redis sliding window counter

Data Validation

javascript

```
// Setiap request divalidasi dengan Joi/Zod sebelum controller
const checkinSchema = z.object({
  method: z.enum(['qr', 'selfie', 'gps']),
  qrToken: z.string().optional(),
  photo: z.string().base64().optional(),
  latitude: z.number().min(-90).max(90),
  longitude: z.number().min(-180).max(180),
  accuracy: z.number().positive().optional()
})
```

SQL Injection Prevention

- Semua query menggunakan parameterized queries (pg driver)
- ORM: Knex.js dengan query builder — no raw string interpolation
- Input sanitization middleware (express-validator)

CORS Configuration

```
javascript

cors({
  origin: process.env.CORS_ORIGINS.split(','),
  credentials: true,
  methods: ['GET', 'POST', 'PATCH', 'DELETE'],
  allowedHeaders: ['Content-Type', 'Authorization']
})
```

File Upload Security

```
Validasi upload foto selfie:
1. MIME type check (hanya image/jpeg, image/png)
2. File size limit: 5MB
3. Scan magic bytes (bukan hanya extension)
4. Resize & strip metadata sebelum store ke R2
5. Generate random filename (bukan nama asli user)
6. URL signed (private) – akses hanya via API, bukan langsung
```

HTTPS & Headers

```
Semua response menggunakan security headers:
- Strict-Transport-Security: max-age=31536000
- X-Content-Type-Options: nosniff
- X-Frame-Options: DENY
- Content-Security-Policy: default-src 'self'
- X-XSS-Protection: 1; mode=block
```

7. Caching Strategy

Redis Cache Keys

Key Pattern	Content	TTL	Invalidate When
qr_current:{locationId}	QR token aktif	35s	Generate baru

Key Pattern	Content	TTL	Invalidate When
<code>qr:{token}</code>	Token payload	35s	Expire auto
<code>qr:{token}:used</code>	Flag used	10s	Auto
<code>org:{orgId}:employees</code>	List karyawan	5 menit	Employee update
<code>org:{orgId}:locations</code>	List lokasi	10 menit	Location update
<code>org:{orgId}:shifts</code>	List shift	10 menit	Shift update
<code>dashboard:{orgId}:{date}</code>	Dashboard stats	2 menit	Checkin event
<code>session:{userId}</code>	Session data	15 menit	Logout
<code>ratelimit:{ip}:{endpoint}</code>	Counter	Per window	Auto

Cache Strategy per Use Case

QR Token (Ultra-short TTL):

- Read-through: Cek Redis dulu, kalau miss generate baru
- Write-through: Generate → save Redis → async save PostgreSQL
- TTL sangat pendek (35 detik) untuk security

Dashboard Stats (Short TTL):

- Cache-aside: Load dari DB, cache di Redis 2 menit
- Invalidate setiap ada check-in baru (cache bust)
- Acceptable stale: 2 menit untuk dashboard

Config Data (Long TTL):

- Org config, lokasi, shift → cache 10 menit
- Invalidate on write: setiap admin update, bust cache

Report Data (No cache):

- Laporan selalu fresh dari DB
- Heavy query: optimize dengan proper indexes + pagination

8. Offline & Sync Strategy

PWA Service Worker

javascript

```
// Cache strategy per resource type:
precacheAndRoute([
  { url: '/', revision: BUILD_HASH },
  { url: '/static/js/main.js', revision: BUILD_HASH },
  // ... semua static assets
])

// Runtime caching:
registerRoute(
  /\api\/attendance\/history/,
  new StaleWhileRevalidate({ cacheName: 'attendance-history' })
)

registerRoute(
  /\api\/admin\/dashboard/,
  new NetworkFirst({
    cacheName: 'dashboard',
    networkTimeoutSeconds: 3
  })
)
```

Offline Queue (IndexedDB)

javascript

```
// Schema offline queue
const offlineQueue = {
  storeName: 'pendingCheckins',
  schema: {
    id: 'auto',
    userId: 'string',
    method: 'string',          // 'gps' only (QR butuh online)
    latitude: 'number',
    longitude: 'number',
    accuracy: 'number',
    photoBlob: 'blob',        // Untuk selfie offline
    submittedAt: 'timestamp', // Waktu asli check-in
    syncedAt: null,
    attempts: 0
  }
}

// Sync saat online kembali
window.addEventListener('online', async () => {
  const queue = await getOfflineQueue()
  for (const item of queue) {
    try {
      await submitCheckin(item)
      await removeFromQueue(item.id)
    } catch (err) {
      await incrementAttempts(item.id)
    }
  }
})
```

Conflict Resolution

Skenario	Resolution
Offline checkin, sudah manual check-in oleh admin	Offline submission ditolak (409 Conflict), log ke audit
2 offline checkin di hari sama	Pertama diterima, kedua ditolak
Offline checkout + sudah ada	Ambil yang lebih logis (waktu lebih awal = check-in, lebih akhir = check-out)

9. Notification System

Event → Notification Mapping

Event	Penerima	Channel	Timing
Check-in berhasil	Karyawan	Push	Immediate
Check-in terlambat	Karyawan + Admin	Push + Email	Immediate
Karyawan tidak hadir	Admin	Push + Email	+30 menit dari jam masuk
Request izin baru	Admin	Push + Email	Immediate
Request disetujui	Karyawan	Push + Email	Immediate
Request ditolak	Karyawan	Push + Email	Immediate
QR Code expired	Karyawan (saat scan)	In-app toast	On event
Check-in offline tersync	Karyawan	Push	On sync
Laporan bulanan	Admin	Email	1 tgl bulan berikutnya
Kuota cuti < 3 hari	Karyawan	Push	Setelah approval
Renewal billing	Superadmin	Email	H-7 renewal

Email Templates (Resend)

Template yang diperlukan:

- | | |
|--------------------------------|-----------------------------|
| 1. otp-verification.html | → Verifikasi OTP registrasi |
| 2. employee-invite.html | → Undangan karyawan baru |
| 3. password-reset.html | → Reset password |
| 4. leave-request-approved.html | → Notifikasi approval |
| 5. leave-request-rejected.html | → Notifikasi rejection |
| 6. monthly-report.html | → Laporan bulanan otomatis |
| 7. billing-renewal.html | → Reminder renewal |

Push Notification (FCM)

javascript

```
// Payload FCM
const notification = {
  token: user.pushToken,
  notification: {
    title: "Check-in Berhasil ✓",
    body: "Kehadiran Anda tercatat pukul 08:02 WIB"
  },
  data: {
    type: "checkin_success",
    attendanceId: "uuid",
    action: "open_home"
  },
  android: {
    notification: { channelId: "attendance" }
  },
  apns: {
    payload: { aps: { sound: "default" } }
  }
}
```

10. Observability & Monitoring

Metrics (Prometheus)

Metrics yang dikumpulkan:

Business metrics:

- hadir_checkins_total{method, status, org}
- hadir_requests_total{type, status}
- hadir_active_orgs_total
- hadir_active_users_total{role}

Technical metrics:

- http_request_duration_seconds{method, route, status}
- http_requests_total{method, route, status}
- db_query_duration_seconds{query}
- redis_cache_hits_total
- redis_cache_misses_total
- queue_size{queue_name}

Alerting Rules

Alert	Condition	Severity
API Error Rate High	error_rate > 5% selama 5 menit	Critical
DB Slow Queries	p99 latency > 1s selama 3 menit	Warning
Redis Down	redis_up == 0	Critical
Check-in Spike	checkins > 3x normal 15 menit	Info
Disk Usage	disk_used > 80%	Warning
QR Token Gen Fail	qr_gen_errors > 0 selama 2 menit	Critical

Logging Strategy

```
javascript

// Structured logging (Winston)
logger.info('checkin_success', {
  userId, orgId, method,
  lat, lng, locationId,
  duration_ms: Date.now() - startTime
})

logger.error('qr_validation_failed', {
  token: token.substring(0, 8) + '...', // Jangan log full token
  reason: 'expired',
  userId
})

// Log levels:
// ERROR: Production issues yang butuh action
// WARN:  Anomali yang tidak blocking
// INFO:  Business events penting
// DEBUG: Development only (disable di production)
```

Health Check Endpoint

GET `/health` (no auth required)

```
json
```

```
Response 200:
{
  "status": "healthy",
  "timestamp": "2026-04-28T01:41:00Z",
  "checks": {
    "database": "healthy",
    "redis": "healthy",
    "storage": "healthy"
  },
  "version": "1.0.0"
}
```

Uptime SLA Target

Tier	Target	Max Downtime/Month
Free	99.0%	7.2 jam
Starter	99.5%	3.6 jam
Pro	99.9%	43 menit
Enterprise	99.95%	21 menit

System Design ini adalah dokumen hidup — update setiap ada keputusan arsitektur signifikan.

Next steps setelah dokumen ini:

1. Setup monorepo (pnpm workspace: (/apps/web), (/apps/api), (/packages/shared))
2. Initialize PostgreSQL schema dengan migrations (Knex)
3. Build auth module dulu (register → OTP → login → JWT)
4. Lalu attendance module (QR + GPS)
5. Baru selfie + admin dashboard